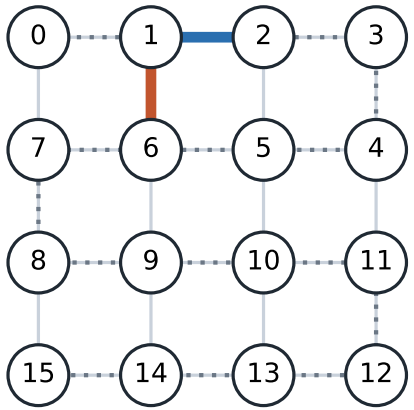
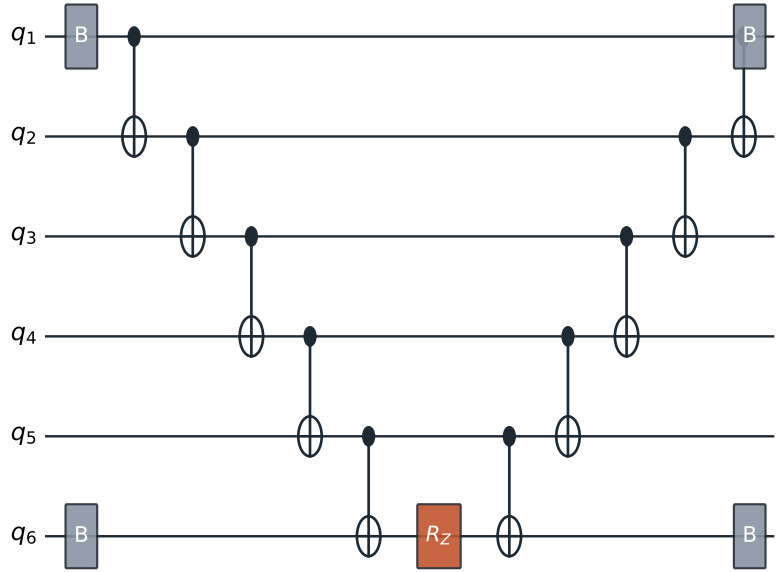
(a) Jordan–Wigner snake, 4×4

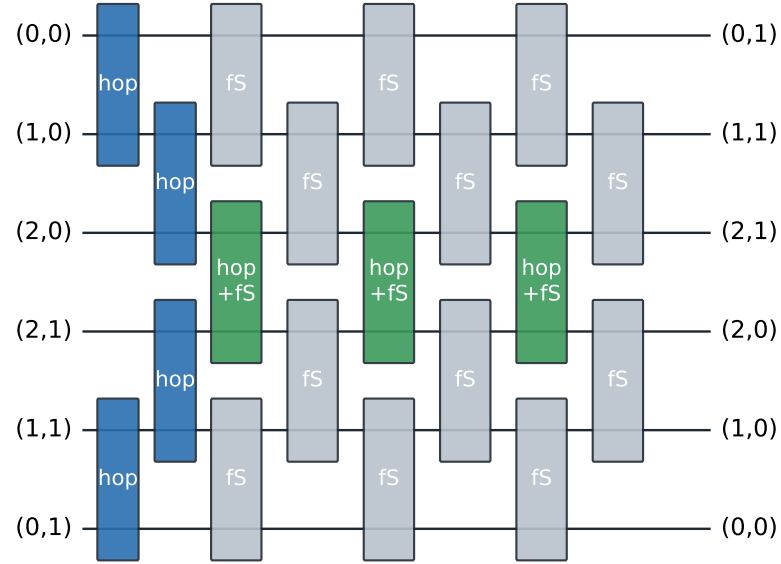


(b) Direct: $e^{-i\theta X_1Z_2\cdots Z_5X_6}$



one Pauli term of the long hop: parity ladder over every qubit between
2(k−1) CNOTs in depth ∝ k, for modes k apart (B: basis change)

(c) Fermionic-swap network, 2×3 strip



odd–even fSWAP layers reverse the order: every pair of modes is adjacent once,
so all 3 vertical hops act on neighbouring qubits; no long Z strings.

fS: fermionic SWAP (SWAP with a −1 phase on |11⟩), which exchanges two adjacent modes and keeps the Jordan–Wigner order consistent. The physical arm uses such networks on the full lattice (Kivlichan et al.), so each Trotter step needs O(N·L) nearest-neighbour two-qubit gates; hence the physical arm's G ∝ rL³ gate count.